3.3 Graduate algebra: study notes

These notes condense the module. Rings are commutative with 1 unless stated otherwise (the group algebra $\mathbb{C}[G]$ in §12 is the exception); ring homomorphisms preserve 1. A denotes a ring, M, N are A-modules, k is a field. Read the texts first; use these notes to consolidate and review.

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1. Categories, functors and universal properties (light)

A **category** $\mathcal{C}$ has objects, sets of morphisms $\mathrm{Hom}(X, Y)$, an associative composition and identities. Examples: Set, Grp, Ab, A-Mod, Ring, Top, and a poset (one morphism $x \to y$ iff $x \le y$).

A (covariant) **functor** $F : \mathcal{C} \to \mathcal{D}$ sends objects to objects and morphisms to morphisms, preserving composition and identities; a contravariant functor reverses arrows. Examples: forgetful functors; $\mathrm{Hom}(M, -)$ (covariant) and $\mathrm{Hom}(-, N)$ (contravariant) from A-Mod to A-Mod; Spec : Ring $\to$ Top (contravariant); homology H_n : Top $\to$ Ab (in 3.4).

A **natural transformation** $\eta : F \Rightarrow G$ is a family $\eta_X : F(X) \to G(X)$ with $G(f) \circ \eta_X = \eta_Y \circ F(f)$ for every $f : X \to Y$. Example: the map $V \to V^{**}$ into the double dual is natural; an isomorphism $V \cong V^*$ is not (it depends on a basis).

Universal properties. An object defined by a universal property is unique up to unique isomorphism. *Product* $X \times Y$: an object with maps p_X, p_Y such that every pair $f : Z \to X$, $g : Z \to Y$ factors uniquely through it. *Coproduct*: the dual notion. In Set these are the cartesian product and disjoint union; in A-Mod, finite products and coproducts coincide (the direct sum $M \oplus N$), but for infinite families $\prod M_i \neq \bigoplus M_i$; in Grp the coproduct is the free product; in commutative rings it is $A \otimes_{\mathbb{Z}} B$.

Adjunctions. $F : \mathcal{C} \to \mathcal{D}$ is left adjoint to $G : \mathcal{D} \to \mathcal{C}$ if $\mathrm{Hom}_{\mathcal{D}}(FX, Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, GY)$ naturally in X and Y. Examples: free module $\dashv$ forgetful; $- \otimes_A M \dashv \mathrm{Hom}_A(M, -)$. *Left adjoints preserve colimits (cokernels, direct sums) and are right exact; right adjoints preserve limits (kernels, products) and are left exact.* This one principle explains the exactness properties in §3.

Yoneda lemma (statement). $X \cong Y$ iff $\mathrm{Hom}(-, X) \cong \mathrm{Hom}(-, Y)$ naturally. "An object is determined by how others map into it."

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2. Modules and exact sequences

Modules, submodules, quotients and the isomorphism theorems work exactly as for abelian groups. M is **free** if it has a basis; every module is a quotient of a free module. **Finitely generated** means a quotient of A^n.

A sequence $\cdots \to M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \to \cdots$ is **exact** at M_i if $\mathrm{im}\, f_i = \ker f_{i+1}$. A **short exact sequence** $0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$ means f injective, g surjective, $\mathrm{im}\, f = \ker g$. So $M'' \cong M/f(M')$.

Splitting lemma. For a short exact sequence as above, the following are equivalent: (a) there is $r : M \to M'$ with $rf = \mathrm{id}$; (b) there is $s : M'' \to M$ with $gs = \mathrm{id}$; (c) there is an isomorphism $M \cong M' \oplus M''$ compatible with f and g. Non-example: $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ does not split ($\mathbb{Z}$ has no element of order 2). Over a field every short exact sequence splits.

Five lemma. In a commutative diagram with exact rows of five terms each, if the outer four vertical maps are isomorphisms, so is the middle one.

Snake lemma. A morphism of short exact sequences $(M' \to M \to M'') \to (N' \to N \to N'')$ with vertical maps α, β, γ gives an exact sequence

$$0 \to \ker \alpha \to \ker \beta \to \ker \gamma \xrightarrow{\delta} \operatorname{coker} \alpha \to \operatorname{coker} \beta \to \operatorname{coker} \gamma \to 0.$$

The *connecting homomorphism* δ is the prototype of the long exact sequences of homology in 3.4. Prove it once by diagram chasing; afterwards use it freely.

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3. Tensor products and Hom

Definition. $M \otimes_A N$ is an A-module with a bilinear map $\otimes : M \times N \to M \otimes_A N$ such that every bilinear $\beta : M \times N \to P$ factors uniquely as $\beta = \tilde{\beta} \circ \otimes$ with $\tilde{\beta}$ linear. Existence: the free module on $M \times N$ modulo the bilinearity relations. Elements are finite sums $\sum m_i \otimes n_i$; **not every element is a pure tensor**, and a pure tensor can be zero for non-obvious reasons.

Basic isomorphisms. $A \otimes_A M \cong M$; $M \otimes N \cong N \otimes M$; $(M \otimes N) \otimes P \cong M \otimes (N \otimes P)$; $(\bigoplus M_i) \otimes N \cong \bigoplus (M_i \otimes N)$; $A^m \otimes A^n \cong A^{mn}$; $(A/I) \otimes_A M \cong M/IM$; $S^{-1}A \otimes_A M \cong S^{-1}M$.

Examples over $\mathbb{Z}$.

- $\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m,n)$.
- $\mathbb{Q} \otimes \mathbb{Q}/\mathbb{Z} = 0$ (divisible ⊗ torsion = 0); $\mathbb{Q} \otimes \mathbb{Q} \cong \mathbb{Q}$.
- $\mathbb{Z}/n \otimes M \cong M/nM$.
- $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ as rings.

Extension of scalars. For a ring map $A \to B$, $B \otimes_A M$ is a B-module. Example: complexification $\mathbb{C} \otimes_{\mathbb{R}} V$. For algebras, $B \otimes_A C$ is a ring with $(b \otimes c)(b' \otimes c') = bb' \otimes cc'$; $A[x] \otimes_A A[y] \cong A[x,y]$, and $k[x]/(f) \otimes_k L \cong L[x]/(f)$.

Hom–tensor adjunction. $\operatorname{Hom}_A(M \otimes_A N, P) \cong \operatorname{Hom}_A(N, \operatorname{Hom}_A(M,P))$, $\varphi \mapsto (n \mapsto (m \mapsto \varphi(m \otimes n)))$.

Exactness.

- $\operatorname{Hom}_A(M, -)$ is **left exact**: if $0 \to N' \to N \to N''$ is exact then so is $0 \to \operatorname{Hom}(M,N') \to \operatorname{Hom}(M,N) \to \operatorname{Hom}(M,N'')$. Similarly $\operatorname{Hom}(-,N)$ turns $M' \to M \to M'' \to 0$ into $0 \to \operatorname{Hom}(M'',N) \to \operatorname{Hom}(M,N) \to \operatorname{Hom}(M',N)$.
- $M \otimes_A -$ is **right exact**: $N' \to N \to N'' \to 0$ exact implies $M \otimes N' \to M \otimes N \to M \otimes N'' \to 0$ exact. *Proof:* a sequence $X' \to X \to X'' \to 0$ is exact iff $0 \to \operatorname{Hom}(X'',P) \to \operatorname{Hom}(X,P) \to \operatorname{Hom}(X',P)$ is exact for every P. Apply this with $X = M \otimes N$ and use the adjunction plus left exactness of $\operatorname{Hom}(-, \operatorname{Hom}(M,P))$.
- Neither is exact in general: $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is injective, but after $\mathbb{Z}/2 \otimes -$ it becomes the zero map $\mathbb{Z}/2 \to \mathbb{Z}/2$. And $\mathbb{Z} \to \mathbb{Z}/2$ is onto, but $\operatorname{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0 \to \operatorname{Hom}(\mathbb{Z}/2, \mathbb{Z}/2)$ is not.

The failure of exactness is measured by the derived functors Tor and Ext (e.g. $\operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m,n)$, $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z}) \cong \mathbb{Z}/n$). They appear in the universal coefficient theorems of 3.4.

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4. Projective, injective and flat modules

Projective: P is projective if $\operatorname{Hom}(P, -)$ is exact; equivalently every surjection $M \to P$ splits; equivalently every map $P \to M''$ lifts through any surjection $M \to M''$; equivalently P is a direct summand of a free module. Free modules are projective. Over a PID (and over local rings, for finitely generated modules) projective = free. Not in general: in $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$, the summand $\mathbb{Z}/2$ is projective but not free.

Injective: I is injective if $\mathrm{Hom}(-, I)$ is exact; equivalently, maps into I extend along injections. **Baer's criterion:** it suffices to extend maps from ideals $\mathfrak{a} \to I$ to $A \to I$. Over a PID, injective = divisible ($nx = y$ solvable for all y and $n \neq 0$). So $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are injective $\mathbb{Z}$-modules, while $\mathbb{Z}$ is not. Every module embeds in an injective module.

Flat: F is flat if $F \otimes_A -$ is exact (preserves injections). Free ⇒ projective ⇒ flat. $\mathbb{Q}$ is a flat $\mathbb{Z}$-module that is not projective. Localisations $S^{-1}A$ are flat over A. Over a PID, flat = torsion-free. Flatness is local: M is flat iff $M_{\mathfrak{p}}$ is flat over $A_{\mathfrak{p}}$ for all primes $\mathfrak{p}$.

Why this matters. Projective and injective modules are used to build resolutions, which compute Ext and Tor; flat base change is the algebraic notion of "a family varying nicely" in algebraic geometry.

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5. Ideals, radicals and Nakayama

Existence of maximal ideals. Every proper ideal is contained in a maximal ideal (Zorn). So $A \neq 0$ has a maximal ideal, and a non-unit lies in some maximal ideal.

Nilradical. $\mathrm{nil}(A) = \{a : a^n = 0 \text{ for some } n\} = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$. *Proof of "⊇":* if f is not nilpotent, $S = \{1, f, f^2, \dots\}$ avoids 0, so $S^{-1}A \neq 0$ has a maximal ideal; its preimage in A is a prime not containing f. More generally $\sqrt{I} = \bigcap_{\mathfrak{p} \supseteq I} \mathfrak{p}$.

Jacobson radical. $J(A) = \bigcap_{\mathfrak{m} \text{ maximal}} \mathfrak{m} = \{a : 1 - ab \text{ is a unit for all } b\}$.

Local rings. A is local if it has a unique maximal ideal $\mathfrak{m}$; equivalently the non-units form an ideal. Examples: fields, $\mathbb{Z}_{(p)}$, $k[[x]]$, $A_{\mathfrak{p}}$.

Nakayama's lemma. Let M be a finitely generated A-module and $I \subseteq J(A)$ an ideal. If $IM = M$ then $M = 0$.

Proof. Suppose $M \neq 0$ with minimal generating set $x_1, \dots, x_n$. Since $x_n \in IM$, $x_n = \sum_i a_i x_i$ with $a_i \in I$. Then $(1 - a_n)x_n = \sum_{i<n} a_i x_i$, and $1 - a_n$ is a unit because $a_n \in J(A)$. So x_n is redundant, a contradiction. □

Corollaries. If $N \subseteq M$, M finitely generated, and $N + IM = M$, then $N = M$. Over a local ring $(A, \mathfrak{m}, k)$, elements $x_1, \dots, x_n$ generate M iff their images span $M/\mathfrak{m}M$ over k; so every minimal generating set has $\dim_k M/\mathfrak{m}M$ elements. Finitely generated projective modules over a local ring are free.

Pitfall: finite generation is essential. $\mathbb{Q}$ over $\mathbb{Z}_{(p)}$ satisfies $p\mathbb{Q} = \mathbb{Q}$ but $\mathbb{Q} \neq 0$.

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6. Localisation

For a multiplicative set $S \ni 1$, $S^{-1}A = \{a/s\}$ with $a/s = b/t$ iff $u(at - bs) = 0$ for some $u \in S$. (The u is needed when A has zero divisors.) Universal property: ring maps $A \to B$ sending S to units factor uniquely through $S^{-1}A$. Similarly $S^{-1}M$; $S^{-1}M \cong S^{-1}A \otimes_A M$.

Key cases. $A_f = S^{-1}A$ for $S = \{f^n\}$ ("invert f"); $A_{\mathfrak{p}} = S^{-1}A$ for $S = A \setminus \mathfrak{p}$ ("look near $\mathfrak{p}$"), a local ring with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$; the fraction field of a domain ($S = A \setminus \{0\}$).

Properties.

- **Exactness:** S^{-1} is an exact functor; equivalently $S^{-1}A$ is flat over A.

- **Primes:** primes of $S^{-1}A$ correspond bijectively to primes of A not meeting S (via $\mathfrak{q} \mapsto \mathfrak{q} \cap A$). So the primes of $A_\mathfrak{p}$ are the primes contained in $\mathfrak{p}$, and the primes of $A/\mathfrak{p}$ are those containing $\mathfrak{p}$.
- **Local properties:** $M = 0$ iff $M_\mathfrak{p} = 0$ for all primes $\mathfrak{p}$ iff $M_\mathfrak{m} = 0$ for all maximal $\mathfrak{m}$. The same holds for injectivity and surjectivity of a map, and for flatness. Being a domain or reduced: reduced is local; domain is not (e.g. $k \times k$).

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7. The prime spectrum

Spec A is the set of prime ideals with the **Zariski topology**, whose closed sets are $V(I) = \{\mathfrak{p} \supseteq I\}$. Basic open sets $D(f) = \operatorname{Spec} A \setminus V(f)$ form a base, and $D(f) \cong \operatorname{Spec} A_f$.

Facts.

- $V(I) \subseteq V(J)$ iff $\sqrt{I} \supseteq \sqrt{J}$; $V(I) = \emptyset$ iff $I = A$.
- The closure of $\{\mathfrak{p}\}$ is $V(\mathfrak{p})$; closed points are maximal ideals.
- Spec A is quasi-compact (open covers have finite subcovers), though rarely Hausdorff.
- A ring map $\varphi : A \to B$ induces a continuous map $\operatorname{Spec} B \to \operatorname{Spec} A$, $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$. $\operatorname{Spec}(A/I) \cong V(I)$ (closed embedding) and $\operatorname{Spec}(S^{-1}A)$ is the subspace of primes missing S.
- Spec A is irreducible iff $\operatorname{nil}(A)$ is prime.

Examples.

- $\operatorname{Spec} \mathbb{Z} = \{(0)\} \cup \{(p)\}$. Closed sets: the whole space and finite sets of non-zero primes. (0) is a dense "generic point".
- $\operatorname{Spec} \mathbb{C}[x] = \{(0)\} \cup \{(x-a) : a \in \mathbb{C}\}$: the complex line plus a generic point.
- $\operatorname{Spec} \mathbb{R}[x]$: (0), $(x-a)$ for $a \in \mathbb{R}$, and (x^2+bx+c) irreducible, i.e. conjugate pairs of non-real points.
- $\operatorname{Spec} \mathbb{C}[x,y]$: (0), irreducible curves (f), and points $(x-a, y-b)$ (by the Nullstellensatz).
- $\operatorname{Spec} \mathbb{Z}[x]$: Mumford's "arithmetic surface": (0), (p), (f) for irreducible primitive f, and maximal ideals (p, f) with f irreducible mod p.

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8. Noetherian rings

For a module M, the following are equivalent: every ascending chain of submodules stabilises (ACC); every non-empty set of submodules has a maximal element; every submodule is finitely generated. M is then **Noetherian**. A ring is Noetherian if it is Noetherian as a module over itself, i.e. every ideal is finitely generated.

If $0 \to M' \to M \to M'' \to 0$ is exact, M is Noetherian iff M' and M'' are. Hence finitely generated modules over a Noetherian ring are Noetherian. Quotients and localisations of Noetherian rings are Noetherian.

Hilbert basis theorem. If A is Noetherian, so is $A[x]$.

Proof. Suppose $I \subseteq A[x]$ is not finitely generated. Choose $f_1 \in I$ of minimal degree, and inductively $f_{k+1} \in I \setminus (f_1, \dots, f_k)$ of minimal degree. Then $\deg f_1 \leq \deg f_2 \leq \cdots$. Let a_k be the leading coefficient of f_k. The chain $(a_1) \subseteq (a_1, a_2) \subseteq \cdots$ stabilises, so $a_{N+1} = \sum_{i \leq N} r_i a_i$ for some N. Then

$$g = f_{N+1} - \sum_{i \leq N} r_i x^{\deg f_{N+1} - \deg f_i} f_i$$

lies in $I \setminus (f_1, \dots, f_N)$ and has degree $< \deg f_{N+1}$, contradicting minimality. □

Corollaries. $k[x_1, \dots, x_n]$ and $\mathbb{Z}[x_1, \dots, x_n]$ are Noetherian; so is every finitely generated algebra over a Noetherian ring. Geometrically, every affine variety is cut out by finitely many polynomials.

Non-examples. $k[x_1, x_2, \dots]$ (infinitely many variables); $C[0,1]$; the ring of algebraic integers $\overline{\mathbb{Z}}$ ($(2) \subsetneq (2^{1/2}) \subsetneq (2^{1/4}) \subsetneq \cdots$). A subring of a Noetherian ring need not be Noetherian ($k[x, xy, xy^2, \dots] \subseteq k[x,y]$).

Related results. In a Noetherian ring every ideal contains a power of its radical, and every ideal has a primary decomposition (Lasker–Noether). **Artinian** rings (DCC) are Noetherian of dimension 0 (AM ch. 8).

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9. Integral extensions

Let $A \subseteq B$. An element $b \in B$ is **integral** over A if it satisfies a monic polynomial with coefficients in A. Equivalently, $A[b]$ is a finitely generated A-module; equivalently, b lies in a subring $C \subseteq B$ that is a finitely generated A-module. (The last $\Rightarrow$ first uses the determinant trick: if $bC \subseteq C = \sum Ac_i$, write $bc_i = \sum a_{ij}c_j$; then $\det(bI - (a_{ij}))$ kills $C \ni 1$.)

Consequences. The integral elements form a subring, the **integral closure** of A in B. Integrality is transitive. A is **integrally closed** (normal) if it equals its integral closure in its fraction field. UFDs are integrally closed (rational root test); so are $\mathbb{Z}$, $k[x_1, \dots, x_n]$, and all localisations of normal domains. $\mathbb{Z}[\sqrt{-3}]$ and $k[x,y]/(y^2 - x^3)$ are not: $\frac{1+\sqrt{-3}}{2}$ and y/x are integral but missing.

Field lemma. If $A \subseteq B$ are domains with B integral over A, then A is a field iff B is a field.

Cohen–Seidenberg theorems (for $A \subseteq B$ integral):

- **Lying over:** for every prime $\mathfrak{p}$ of A there is a prime $\mathfrak{q}$ of B with $\mathfrak{q} \cap A = \mathfrak{p}$. (Localise at $\mathfrak{p}$: $A_\mathfrak{p} \subseteq B_\mathfrak{p}$ is still integral; take any maximal ideal $\mathfrak{n}$ of $B_\mathfrak{p}$; by the field lemma $\mathfrak{n} \cap A_\mathfrak{p}$ is maximal, hence $\mathfrak{p}A_\mathfrak{p}$; pull back to B.)
- **Incomparability:** if $\mathfrak{q} \subseteq \mathfrak{q}'$ lie over the same $\mathfrak{p}$ then $\mathfrak{q} = \mathfrak{q}'$.
- **Going up:** chains of primes in A lift to chains in B extending a given start.
- Hence $\dim A = \dim B$ (Krull dimension: the supremum of lengths of prime chains).

Example. $\mathbb{Z} \subseteq \mathbb{Z}[i]$ is integral. Over (2) lies $(1+i)$ (ramified: $(2) = (1+i)^2$); over $p \equiv 1 \pmod 4$ lie two primes $(a+bi)$, $(a-bi)$ with $p = a^2 + b^2$ (split); over $p \equiv 3 \pmod 4$ lies one prime (p) (inert).

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10. Noether normalisation and the Nullstellensatz

Noether normalisation. If B is a finitely generated k-algebra, there are algebraically independent $y_1, \dots, y_d \in B$ such that B is finite (integral and finitely generated as a module) over $k[y_1, \dots, y_d]$. Moreover $d = \dim B$. (Proof by a change of variables $x_i \mapsto x_i - x_n^{r_i}$ that makes a relation monic in x_n, then induction.)

Zariski's lemma. If a finitely generated k-algebra L is a field, then L is a finite extension of k. (By normalisation L is integral over $k[y_1, \dots, y_d]$; by the field lemma the polynomial ring is a field, so $d = 0$.)

Weak Nullstellensatz. If k is algebraically closed, the maximal ideals of $k[x_1, \dots, x_n]$ are exactly $\mathfrak{m}_a = (x_1 - a_1, \dots, x_n - a_n)$ for $a \in k^n$. Equivalently, if $I \subsetneq k[x_1, \dots, x_n]$ is a proper ideal then $V(I) = \{a \in k^n : f(a) = 0 \ \forall f \in I\} \neq \emptyset$.

Proof. If $\mathfrak{m}$ is maximal, $L = k[x]/\mathfrak{m}$ is a field, finite over k by Zariski, hence equal to k. Let $a_i \in k$ be the image of x_i. Then $\mathfrak{m}_a \subseteq \mathfrak{m}$, and $\mathfrak{m}_a$ is maximal, so they are equal. $\square$

(For $k = \mathbb{C}$ there is a slicker proof: L has countable dimension over $\mathbb{C}$, but if $t \in L$ were transcendental, the uncountably many elements $1/(t-a)$, $a \in \mathbb{C}$, would be linearly independent. See Problem 18.)

Strong Nullstellensatz. For k algebraically closed and any ideal I, $I(V(I)) = \sqrt{I}$.

Proof (Rabinowitsch trick). Let f vanish on $V(I)$, $I = (g_1, \dots, g_m)$ (Hilbert basis theorem). In $k[x_1, \dots, x_n, t]$, the ideal $(g_1, \dots, g_m, 1 - tf)$ has no common zero, so by the weak form it is the whole ring: $1 = \sum h_i g_i + h(1 - tf)$. Substitute $t = 1/f$ in $k(x)$ and clear denominators: $f^N = \sum \tilde{h}_i g_i \in I$. $\square$

Dictionary. For k algebraically closed, V and I give inclusion-reversing bijections:

Algebra ($k[x_1, \dots, x_n]$)	Geometry (k^n)
radical ideals	affine algebraic sets
prime ideals	irreducible algebraic sets (varieties)
maximal ideals	points
$k[x]/I$ (reduced)	regular functions on $V(I)$

Also: a finitely generated k-algebra is a **Jacobson ring**: every prime is an intersection of maximal ideals, so the nilradical equals the Jacobson radical.

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11. Dedekind domains

A **discrete valuation ring** (DVR) is a local PID that is not a field; equivalently, a Noetherian local domain of dimension 1 that is integrally closed. Every non-zero ideal is (π^n) for a uniformiser π. Examples: $\mathbb{Z}_{(p)}$, $k[[t]]$, $k[x]_{(x)}$.

A **Dedekind domain** is a Noetherian, integrally closed domain of dimension ≤ 1. Equivalently: a Noetherian domain whose localisations at non-zero primes are DVRs. Examples: PIDs; rings of integers $\mathcal{O}_K$ of number fields (they are Noetherian as finitely generated $\mathbb{Z}$-modules, integrally closed by definition, and of dimension 1 by going up over $\mathbb{Z}$); coordinate rings of smooth affine curves.

Main theorem. In a Dedekind domain every non-zero ideal factors uniquely as a product of prime ideals. The fractional ideals form a group; modulo principal fractional ideals this is the **ideal class group** $\mathrm{Cl}(A)$, trivial iff A is a PID iff A is a UFD. For rings of integers, Cl is finite (Minkowski).

Rings of integers of quadratic fields. For squarefree $d \neq 0, 1$, the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{d})$ is $\mathbb{Z}[\sqrt{d}]$ if $d \equiv 2, 3 \pmod 4$ and $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ if $d \equiv 1 \pmod 4$.

The classic example. In $\mathbb{Z}[\sqrt{-5}]$, $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ are two factorisations into irreducibles, so it is not a UFD. As ideals,

$$(2) = \mathfrak{p}^2, \quad (3) = \mathfrak{q}\mathfrak{q}', \quad (1 + \sqrt{-5}) = \mathfrak{p}\mathfrak{q}, \quad (1 - \sqrt{-5}) = \mathfrak{p}\mathfrak{q}',$$

with $\mathfrak{p} = (2, 1 + \sqrt{-5})$, $\mathfrak{q} = (3, 1 + \sqrt{-5})$, $\mathfrak{q}' = (3, 1 - \sqrt{-5})$. Both factorisations of 6 are the same ideal factorisation $\mathfrak{p}^2\mathfrak{q}\mathfrak{q}'$, grouped differently. $\mathfrak{p}$ is not principal; $\mathrm{Cl} \cong \mathbb{Z}/2$.

How primes split. For $K = \mathbb{Q}(\sqrt{d})$ and an odd prime $p \nmid d$, p splits if d is a non-zero square mod p and is inert otherwise; primes dividing the discriminant ramify. This is where quadratic reciprocity lives.

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12. Representations of finite groups

Throughout, G is a finite group and representations are on finite-dimensional complex vector spaces.

A **representation** is a homomorphism $\rho : G \to GL(V)$; equivalently a $\mathbb{C}[G]$-module. A **subrepresentation** is a G-invariant subspace; V is **irreducible** if it has no subrepresentations other than 0 and V. Morphisms are **G-maps** (intertwiners). Constructions: direct sums, tensor products ($g(v\otimes w) = gv\otimes gw$), duals, $\mathrm{Hom}(V,W)$ with $(g\varphi)(v) = g\varphi(g^{-1}v)$. The **regular representation** is $\mathbb{C}[G]$ acting on itself.

Maschke's theorem. Every subrepresentation $W \subseteq V$ has a G-invariant complement. Hence every representation is a direct sum of irreducibles.

Proof. Let $p : V \to W$ be any linear projection onto W. Average:

$$\bar{p} = \frac{1}{|G|}\sum_{g\in G} gpg^{-1}.$$

Then $\bar{p}$ is a G-map, maps V into W, and is the identity on W (for $w \in W$, $g^{-1}w \in W$, so $gpg^{-1}w = w$). So $\bar{p}$ is a G-equivariant projection onto W and $\ker\bar{p}$ is an invariant complement. □

The proof needs $|G|$ invertible in the field; in characteristic $p \mid |G|$ Maschke fails (modular representation theory). Over $\mathbb{C}$ one can also average an inner product to make ρ unitary and take orthogonal complements; this version extends to compact groups.

Schur's lemma. If V, W are irreducible and $\varphi : V \to W$ is a G-map, then φ is 0 or an isomorphism (kernel and image are subrepresentations). Over $\mathbb{C}$, every G-map $V \to V$ is a scalar (it has an eigenvalue λ, and $\varphi - \lambda$ is a non-injective G-map). Consequences: irreducible representations of abelian groups are 1-dimensional; the centre acts by scalars on an irreducible representation.

Characters. $\chi_V(g) = \operatorname{tr}\rho(g)$. Properties: $\chi_V(1) = \dim V$; χ_V is a class function; $\chi_{V\oplus W} = \chi_V + \chi_W$; $\chi_{V\otimes W} = \chi_V\chi_W$; $\chi_{V^*} = \overline{\chi_V}$, and $\chi_V(g^{-1}) = \overline{\chi_V(g)}$ because $\rho(g)$ has finite order, so its eigenvalues are roots of unity. The regular character is $|G|$ at 1 and 0 elsewhere.

On class functions define

$$\langle \alpha, \beta\rangle = \frac{1}{|G|}\sum_{g\in G}\alpha(g)\overline{\beta(g)}.$$

Key computation. For any representation U, the averaging map $e = \frac{1}{|G|}\sum_g \rho_U(g)$ is a projection onto the invariants U^G, so $\dim U^G = \operatorname{tr} e = \frac{1}{|G|}\sum_g \chi_U(g)$. Apply this to $U = \mathrm{Hom}(V,W)$, whose character is $\overline{\chi_V}\chi_W$ and whose invariants are the G-maps:

$$\dim \mathrm{Hom}_G(V,W) = \langle \chi_W, \chi_V\rangle.$$

First orthogonality relation. The irreducible characters are orthonormal: $\langle \chi_i, \chi_j\rangle = \delta_{ij}$ (by Schur, $\dim \mathrm{Hom}_G(V_i, V_j) = \delta_{ij}$).

Consequences.

- A representation is determined up to isomorphism by its character. The multiplicity of V_i in V is $\langle \chi_V, \chi_i\rangle$.
- V is irreducible iff $\langle \chi_V, \chi_V\rangle = 1$.
- $\mathbb{C}[G] \cong \bigoplus_i V_i^{\oplus \dim V_i}$, so $\sum_i (\dim V_i)^2 = |G|$.

- **The irreducible characters form a basis of the class functions**, so the number of irreducible representations equals the number of conjugacy classes. (If a class function f is orthogonal to every χ_i, then $\sum_g \overline{f(g)}\rho(g)$ is a G-map on each V_i, hence a scalar, with trace $|G|\langle \chi_i, f\rangle = 0$, hence zero; applying it to $1 \in \mathbb{C}[G]$ gives $f = 0$.)

Second (column) orthogonality. $\sum_i \chi_i(g)\overline{\chi_i(h)} = |C_G(g)|$ if g, h are conjugate and 0 otherwise.

Further facts (statements). $\dim V_i$ divides $|G|$ (via algebraic integers). **Artin–Wedderburn:** $\mathbb{C}[G] \cong \prod_i M_{d_i}(\mathbb{C})$. **Burnside's theorem:** groups of order $p^a q^b$ are solvable (proved with character theory). The one-dimensional representations are the homomorphisms $G \to \mathbb{C}^\times$, which factor through $G/[G,G]$; there are $[G : [G,G]]$ of them.

Induction. For $H \le G$ and a representation W of H, $\mathrm{Ind}_H^G W = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$, of dimension $[G : H] \dim W$, with character

$$\chi_{\mathrm{Ind}\, W}(g) = \frac{1}{|H|} \sum_{x \in G,\ x^{-1}gx \in H} \chi_W(x^{-1}gx).$$

Frobenius reciprocity: $\mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) \cong \mathrm{Hom}_H(W, \mathrm{Res}_H^G V)$, i.e. $\langle \mathrm{Ind}\, \chi_W, \chi_V\rangle_G = \langle \chi_W, \mathrm{Res}\, \chi_V\rangle_H$. (This is the Hom–tensor adjunction of §3 again.)

Character tables.

S_3 (classes: identity; 3 transpositions; 2 three-cycles):

	1	(12)	(123)
trivial	1	1	1
sign	1	-1	1
standard	2	0	-1

A_4 (classes: identity; 3 double transpositions; 4 elements conjugate to (123); 4 conjugate to (132); $\omega = e^{2\pi i/3}$):

	1	$(12)(34)$	(123)	(132)
χ_1	1	1	1	1
χ_2	1	1	ω	ω^2
χ_3	1	1	ω^2	ω
χ_4	3	-1	0	0

(The linear characters come from $A_4/V_4 \cong \mathbb{Z}/3$; χ_4 is forced by $1+1+1+d^2 = 12$ and column orthogonality; it is the restriction of the 3-dimensional rotation representation of the tetrahedron.)

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13. Common pitfalls

1. **Treating every element of $M \otimes N$ as a pure tensor**, or defining a map on $M \otimes N$ by a formula on pure tensors without checking bilinearity.
2. **Forgetting which ring you tensor over.** $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C} \cong \mathbb{C}$ but $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}^2$.

3. **Assuming tensoring preserves injections.** It does only for flat modules.
4. **Dropping the extra u in the definition of localisation** when A has zero divisors.
5. **Applying Nakayama to modules that are not finitely generated.**
6. **Confusing "integral" (algebraic with monic equation over a ring) with "algebraic".** $\frac{1}{2}$ is algebraic over $\mathbb{Z}$ but not integral.
7. **Using the Nullstellensatz over non-algebraically-closed fields.** $(x^2+1) \subseteq \mathbb{R}[x]$ is maximal and $V(x^2+1) = \emptyset$ in $\mathbb{R}$.
8. **Assuming the character table determines the group.** D_4 and Q_8 have the same table (Problem 20).
9. **Using Maschke in characteristic dividing $|G|$.** $\mathbb{Z}/p$ acting on $\mathbb{F}_p^2$ by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has an invariant line with no invariant complement.

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14. Connections forward

- **3.4 Algebraic topology:** chain complexes are sequences of modules; the snake lemma gives long exact sequences in homology; Tor and Ext appear in the universal coefficient and Künneth theorems; homology is a functor and the Eilenberg–Steenrod axioms are categorical.
- **Algebraic geometry:** Spec, localisation and the Nullstellensatz are the foundations of schemes; Noether normalisation gives dimension theory; flatness describes families; Dedekind domains are the algebra of smooth curves.
- **Algebraic number theory:** rings of integers, splitting of primes, class groups, and the finiteness of the class number; local fields via completion of DVRs.
- **3.5 Smooth manifolds and Lie theory:** tensor products and exterior powers of vector spaces become tensor fields and differential forms; representation theory of compact Lie groups (Peter–Weyl) extends §12, with averaging over G replaced by integration against Haar measure (compare 3.1).
- **3.2 Functional analysis:** Schur's lemma and orthogonality reappear for unitary representations on Hilbert spaces; the group algebra $\mathbb{C}[G]$ becomes $L^1(G)$ and the C*-algebra of G.